QP CODE: 25900648

SCHEME OF VALUATION

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

THIRD SEMESTER

Discipline Specific Core Course (DSC) – MG3DSCSTA200

STATISTICAL DISTRIBUTIONS

2024 ADMISSION ONWARDS

Duration: 2 Hours **Max Marks: 70**

Part A Answer any 10 questions (Each question carries 3 marks)

1. For stating $f(x_i) \geq 0$ and $\sum_{i=1}^{\infty} f(x_i) = 1$ -1 Mark
   C=$\frac{1}{10}$ - 2 Marks
2. Definition. -1 Mark
   Expression for mean. -1 Mark
   Expression for variance. - 1 Mark
3. Definition. -3 Mark
4. Definition of mathematical expectation - 2 Marks
   Calculation of mean - 1Mark
5. CLT explanation or statement - 3 Marks
6. Statting the pdf $f(x) = \frac{1}{6} \quad -3 \leq x \leq 3$ - 2 Marks
   Required probability =$\frac{2}{3}$ -1 Mark
7. Definition of standard Laplace. – 1.5 Marks
   Definition of two parameter Laplace – 1.5 Marks
8. Definition of m.g.f – 1 mark
   $M_X(t) = \frac{pe^t}{(1-qe^t)}$ - 2 Marks
9. Any two applications - 3 Marks
10. Student's t statistic - 1 mark
    p.d.f - 2 Marks
11. a = 9.32, b = 14.68 - 3 Marks
12. Finding ʎ = 2 - 1 Mark
    Pdf of Poisson distribution - 1 Mark
    P(X=0) = 0.135 - 1 Mark

Part B Answer any 4 questions (Each question carries 7 marks)

13. Relation between c.d.f and p.d.f - 2 Marks
    Definition of c.d.f - 2 marks

$F(x) = \frac{x}{\theta}$ - 3 Marks

14. Definition of characteristic function -1 Mark
    Statement and proof of each property -2 Marks each
15. State property - 3 Marks

    Proof - 4 Marks

16. For stating the Jacobian formula of change of variables - 1 Mark
    For finding the 2 distributions of Y – 3 Marks each
17. Statement of additive property – 3 Marks
    Proof – 4 Marks
18. Find n, p and q - 2 Marks
    Pdf of Binomial distribution - 2 Marks
    Find p (X ≥ 3) = 0.68 - 3 Marks

Part C Answer any 1 question (Each question carries 12 marks)

19. (a) Definition of F-statistic – 2 Marks
        p.d.f -2 Marks
        Two uses -2 Marks each
    (b) proofs - 2 Marks each
20. For finding $\bar{x} = 79.945$ and SD =5.545 -2 Marks each
    Expected frequencies:

Class	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100
Frequency	3	31	148	322	319	144	30	3

1 mark each for each theoretical frequency -8 Marks